

RAG evaluation

While traditional LLM evaluation focuses on the model's inherent capabilities, RAG evaluation requires a more comprehensive approach that considers both the model's generative abilities and its interaction with external information sources.

RAG systems combine the strengths of LLMs with information retrieval mechanisms, allowing them to generate responses that are not only coherent and contextually appropriate but also grounded in up-to-date, externally sourced information. This makes RAG particularly valuable in fields where current and accurate information is crucial, such as news reporting, research, and customer support.

The evaluation of RAG systems goes beyond assessing a standalone LLM. It requires examining the entire system's performance, including:

- **Retrieval accuracy:** How well does the system fetch relevant information?
- **Integration quality:** How effectively is the retrieved information incorporated into the generated response?
- **Factuality and relevance:** Does the final output address the query appropriately while seamlessly blending retrieved and generated content?

Key metrics for RAG evaluation include retrieval precision and recall, which measure the accuracy and comprehensiveness of the retrieved information. Additionally, the quality of integration between retrieved data and generated text is crucial, as is the overall factuality and coherence of the output.

To illustrate how these metrics are applied in practice, consider a RAG system designed for a customer support chatbot in an e-commerce setting. In this scenario, the user asks “What’s your return policy for laptops purchased during the holiday sale?” The RAG pipeline finds relevant documents on the electronics return policy and documents on holiday sale terms. This additional context is appended at the end of the question, and the model uses it to respond:

```
For laptops purchased during our holiday sale, you have an extended return
period of 60 days from the date of purchase. This is longer than our
standard 30-day return policy for electronics. Please ensure the laptop is
in its original packaging with all accessories to be eligible for a full
refund.
```

Table 7.3: Example of output from a RAG pipeline designed for customer support

In this pipeline, we can evaluate if the retrieved documents correspond to what was expected (retrieval accuracy). We can also measure the difference between responses with and without additional context (integration quality). Finally, we can assess whether the output is relevant and grounded in the information provided by the documents (factuality and relevance).

In this section, we will cover two methods to evaluate how well RAG models incorporate external information into their responses.

Ragas

Retrieval-Augmented Generation Assessment (Ragas) is an open-source toolkit designed to provide developers with a comprehensive set of tools for RAG evaluation and optimization. It’s designed around the idea of **metrics-driven development (MDD)**, a product development approach that relies on data to make well-informed decisions, involving the ongoing monitoring of essential metrics over time to gain valuable insights into an application’s performance. By embracing this methodology, Ragas enables developers to objectively assess their RAG systems, identify areas for improvement, and track the impact of changes over time.

One of the key capabilities of Ragas is its ability to synthetically generate diverse and complex test datasets. This feature addresses a significant pain point in RAG development, as manually creating hundreds of questions, answers, and contexts is both time-consuming and labor-intensive. Instead, it uses an evolutionary approach paradigm inspired by works like Evol-Instruct to craft questions with varying characteristics such as reasoning complexity, conditional elements, and multi-context requirements. This approach ensures a comprehensive evaluation of different components within the RAG pipeline.

Additionally, Ragas can generate conversational samples that simulate chat-based question-and-follow-up interactions, allowing developers to evaluate their systems in more realistic scenarios.

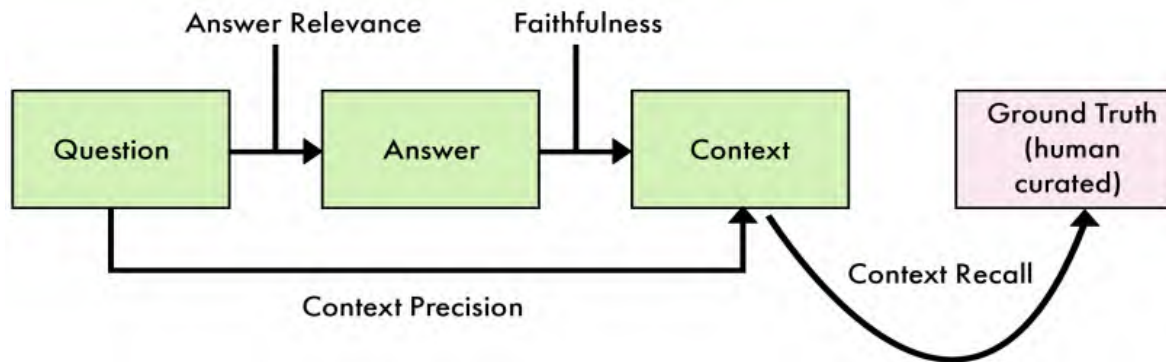


Figure 7.1: Overview of the Ragas evaluation framework

As illustrated in *Figure 7.1*, Ragas provides a suite of LLM-assisted evaluation metrics designed to objectively measure different aspects of RAG system performance. These metrics include:

- **Faithfulness:** This metric measures the factual consistency of the generated answer against the given context. It works by breaking down the answer into individual claims and verifying if each claim can be inferred from the provided context. The faithfulness score is calculated as the ratio of verifiable claims to the total number of claims in the answer.
- **Answer relevancy:** This metric evaluates how pertinent the generated answer is to the given prompt. It uses an innovative approach where an LLM is prompted to generate multiple questions based on the answer and then calculates the mean cosine similarity between these generated questions and the original question. This method helps identify answers that may be factually correct but off-topic or incomplete.
- **Context precision:** This metric evaluates whether all the ground-truth relevant items present in the contexts are ranked appropriately. It considers the position of relevant information within the retrieved context, rewarding systems that place the most pertinent information at the top.
- **Context recall:** This metric measures the extent to which the retrieved context aligns with the annotated answer (ground truth). It analyzes each claim in the ground truth answer to determine whether it can be attributed to the retrieved context, providing insights into the completeness of the retrieved information.

Finally, Ragas also provides building blocks for monitoring RAG quality in production environments. This facilitates continuous improvement of RAG systems. By leveraging the evaluation results from test datasets and insights gathered from production monitoring, developers can iteratively enhance their applications. This might involve fine-tuning retrieval algorithms, adjusting prompt engineering strategies, or optimizing the balance between retrieved context and LLM generation.

Ragas can be complemented with another approach, based on custom classifiers.

ARES

ARES (an automated evaluation framework for RAG systems) is a comprehensive tool designed to evaluate RAG systems. It offers an automated process that combines synthetic data generation with fine-tuned classifiers to assess various aspects of RAG performance, including context relevance, answer faithfulness, and answer relevance.

The ARES framework operates in three main stages: synthetic data generation, classifier training, and RAG evaluation. Each stage is configurable, allowing users to tailor the evaluation process to their specific needs and datasets.

In the synthetic data generation stage, ARES creates datasets that closely mimic real-world scenarios for robust RAG testing. Users can configure this process by specifying document file paths, few-shot prompt files, and output locations for the synthetic queries. The framework supports various pre-trained language models for this task, with the default being google/flan-t5-xxl. Users can control the number of documents sampled and other parameters to balance between comprehensive coverage and computational efficiency.

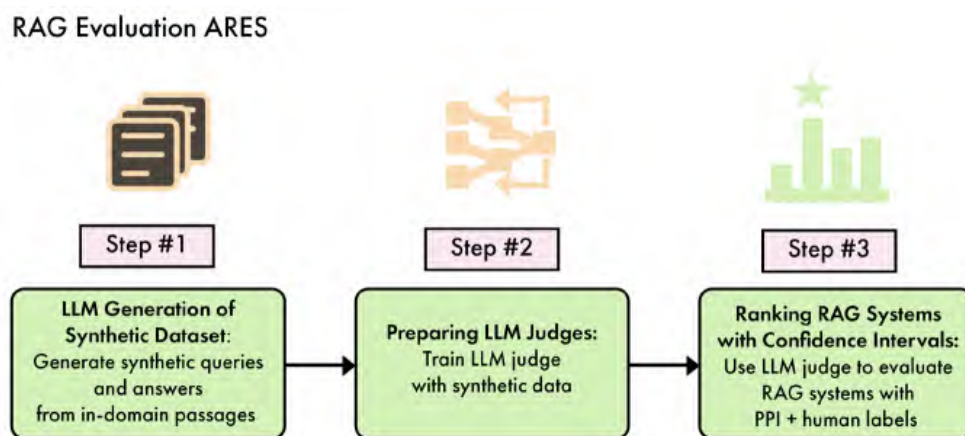


Figure 7.2: Overview of the ARES evaluation framework